

Lower Bounding Covering Size

$$\mathcal{D}(n) = \{(A, B) \in 2^{[n]} \times 2^{[n]} : A \cap B = \emptyset\}$$

Ext. Compl. of $\mathcal{P}_{\text{corr}} \geq$ size of min covering of $\mathcal{D}(n)$ by valid sets.

Lemma: $|\mathcal{D}(n)| = 3^n$

Lemma: Any covering of $\mathcal{D}(n)$ by valid sets has at least 1.5^n ~~max~~ ^{sets}.

We show this by demonstrating that the largest cardinality of any valid subset of $\mathcal{D}(n)$ is at most 2^n .

Thus: $\mathcal{D}(n) \subseteq R_1 \cup R_2 \cup \dots \cup R_r$

$\Rightarrow r \geq 1.5^n$.

Proof — 1/2

Let $p(n)$ = largest card of any valid subset of $\mathcal{D}(n)$.
we will show $p(n) \leq 2^n$.

We prove by induction: $p(n) \leq 2p(n-1) \quad \forall n \geq 1$.

Let $R \subseteq \mathcal{D}(n)$

Define: $R_1 = \left(\{(A, B) \in R : n \in A\} \cup \{(A, B) \in R : (A \cup \{n\}, B) \notin R\} \right)$
 $\cap ([n] \times [n-1])$

$R_2 = \left(\{(A, B) \in R : n \in B\} \cup \{(A, B) \in R : (A, B \cup \{n\}) \notin R\} \right) \cap ([n-1] \times [n])$

Define: $\pi_n : R \rightarrow \mathcal{D}(n-1)$

$(A, B) \rightarrow (A \setminus \{n\}, B \setminus \{n\})$

R valid $\Rightarrow R_1$ valid follows since $R_1 \subseteq R$ similarly R_2

Claim: R_1 valid $\Rightarrow \pi_n(R_1)$ valid.

R_1 valid: $(A, B), (A', B') \in R_1 \Rightarrow |A \cap B'| \neq 1$.

$R_1 \subseteq [n] \times [n-1] \Rightarrow n \notin B' \Rightarrow |A \cap B'| \neq 1 \Rightarrow \pi_n(R_1)$ valid
similarly $\pi_n(R_2)$ valid.

Proof — 2/2

So: $\pi_n(R_1), \pi_n(R_2)$ valid.

By definition of R_1 and R_2 , π_n is injective on R_1 & R_2 : if $(A, B) \in [n-1] \times [n-1]$ with

$(A, B) \in \pi_n(R_1)$: then either $(A \cup n, B) \in R_1$

or $(A, B) \in R_1$, but both cannot happen.

Since π_n is injective on R_1 & R_2 :

$$|R_1| + |R_2| = |\pi_n(R_1)| + |\pi_n(R_2)| \leq 2p^{(n-1)}$$

$$|R| \leq 2p^{(n-1)} \leq 2^n.$$

Hence lemma is proved: the largest valid

subset of $\mathcal{Q}(n)$ has card $\leq 2^n$.